Week 5 | **Exercises** p. 247 {6, 8, 16, 40} | **Assessed** Midterm + Final | **File** 3 / 11

4-2 The Addition Rules for Probability

قواعد الجمع للاحتمال

§4-2 answers exactly one question: **what is P(A *or* B)?** There are two formulas, and choosing between them depends on a single yes/no test.

1 · Mutually exclusive events — the test

Two events are said to be **mutually exclusive events** (or **disjoint**) if they **cannot occur at the same time** — that is, they have no elements in common.

THE TEST

$$A \text{ and } B \text{ are mutually exclusive} \iff A \cap B = \varnothing$$

$\varnothing$ is the null (empty) set. On a Venn diagram, mutually exclusive events are drawn as two circles that **do not overlap**.

الشرح بالعربي — الأحداث المتنافية

الحدثان المتنافيان (**mutually exclusive** أو **disjoint**) هما حدثان **لا يمكن أن يقعا في نفس الوقت**، أي لا يوجد بينهما أي عنصر مشترك: $A \cap B = \varnothing$. على مخطط فن تكون الدائرتان منفصلتين تمامًا.

مثال: عند رمي نرد، «عدد زوجي» $\{2, 4, 6\}$ و«عدد فردي» $\{1, 3, 5\}$ متنافيان. أما «عدد زوجي» و«عامل من عوامل 4» $\{1, 2, 4\}$ فغير متنافيين لأن 2 و 4 مشتركان.

طريقة الإجابة الكاملة: اكتب التقاطع صراحةً ثم احكم — لا تكتفِ بـ «نعم» أو «لا».

WORKED EXAMPLE · ROLLING ONE DIE

S = {1, 2, 3, 4, 5, 6}. A = getting an even number = {2, 4, 6}; B = getting an odd number = {1, 3, 5}; C = getting a factor of 4 = {1, 2, 4}. (a) Are A and B mutually exclusive? (b) Are A and C mutually exclusive?

1. **(a)** $A \cap B = \{2,4,6\} \cap \{1,3,5\} = \varnothing$. **Yes — mutually exclusive.** A number cannot be even and odd at once.
2. **(b)** $A \cap C = \{2,4,6\} \cap \{1,2,4\} = \{2, 4\} \neq \varnothing$. **No — not mutually exclusive.** Rolling a 2 or a 4 satisfies both events simultaneously.

Method mark: always write the intersection out before you answer. "Yes/No" alone is worth less than "$A \cap C = \{2, 4\} \neq \varnothing$, therefore not mutually exclusive."

EXAMPLE 4-15 · DETERMINING MUTUALLY EXCLUSIVE EVENTS

Determine whether the two events are mutually exclusive, and explain your answer.
(a) Randomly selecting a female student / randomly selecting a student who is a second year.
(b) Randomly selecting a person with type A blood / randomly selecting a person with type O blood.
(c) Rolling a die and getting an odd number / rolling a die and getting a number less than 3.
(d) Randomly selecting a person who is under 21 years of age / randomly selecting a person who is over 30 years of age.

1. **(a)** **Not mutually exclusive.** A student can be female *and* a second year at the same time.
2. **(b)** **Mutually exclusive.** A person has exactly one blood type; being type A excludes being type O.
3. **(c)** **Not mutually exclusive.** Odd = {1, 3, 5}; less than 3 = {1, 2}. They share the outcome 1.
4. **(d)** **Mutually exclusive.** No one is simultaneously under 21 and over 30.

2 · The two addition rules

ADDITION RULE #1 — A AND B MUTUALLY EXCLUSIVE

$$P(A \cup B) = P(A) + P(B)$$

ADDITION RULE #2 — A AND B NOT MUTUALLY EXCLUSIVE

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

The overlap gets counted once in P(A) and once again in P(B), so it must be subtracted once to be counted correctly. On the Venn diagram this is the shaded lens in the middle.

الشرح بالعربي — قاعدتا الجمع

قاعدة الجمع تجيب عن سؤال واحد: **ما احتمال A أو B؟**

- **إذا كان الحدثان متنافيين:** نجمع مباشرة `P(A) + P(B)` .
- **إذا كانا غير متنافيين:** نجمع ثم **نطرح التقاطع مرة واحدة**: `P(A) + P(B) − P(A ∩ B)` ، لأن المنطقة المشتركة حُسبت مرتين — مرة داخل A ومرة داخل B.

مثال: 8 ممرضين و 5 أطباء؛ منهم ممرض واحد ذكر وطبيبان ذكور. احتمال (ممرض أو ذكر) = `8/13 + 3/13 − 1/13 = 10/13` ، لأن الممرض الذكر دخل في العدّ مرتين.

USE RULE #2 ALWAYS — IT IS SAFE

Rule #1 is a special case of Rule #2, because when A and B are mutually exclusive, `P(A ∩ B) = 0` and the subtraction disappears. If you are ever unsure under exam pressure, write Rule #2 and show that the intersection is zero. You cannot be wrong, and you gain a method mark for justifying it.

EXAMPLE 4-17 · ENDANGERED SPECIES

In the United States there are 59 species of mammals that are endangered, 75 species of birds, and 68 species of fish. If one animal is selected at random, find the probability that it is either a mammal or a fish.

1. **Total:** n(S) = 59 + 75 + 68 = 202.
2. **Mutually exclusive?** An animal cannot be both a mammal and a fish → **yes**, so use Rule #1.
3. $P(\text{mammal} \cup \text{fish}) = \frac{59}{202} + \frac{68}{202} = \frac{127}{202} \approx$ **0.629**

The 75 birds appear *only* in the denominator. A classic distractor: students who ignore the birds get 127/127 = 1.

EXAMPLE 4-18 · RESEARCH AND DEVELOPMENT EMPLOYEES

Corporate research and development centers for three local companies have these numbers of employees: U.S. Steel 110, Alcoa 750, Bayer Material Science 250. If a research employee is selected at random, find the probability that the employee is employed by U.S. Steel or Alcoa.

1. **Total:** n(S) = 110 + 750 + 250 = 1110.
2. **Mutually exclusive?** An employee works for one company → **yes**, Rule #1.
3. $P(\text{Steel} \cup \text{Alcoa}) = \frac{110}{1110} + \frac{750}{1110} = \frac{860}{1110} = \frac{86}{111} \approx$ **0.775**

EXAMPLE 4-21 · SELECTING A MEDICAL STAFF PERSON

In a hospital unit there are 8 nurses and 5 physicians; 7 nurses and 3 physicians are females. If a staff person is selected, find the probability that the subject is a nurse **or** a male.

Build the two-way table first — always.

STAFF	FEMALE	MALE	TOTAL
Nurses	7	1	8
Physicians	3	2	5
Total	**10**	**3**	**13**

1. Male nurses = 8 − 7 = 1. Male physicians = 5 − 3 = 2. Total males = 3. n(S) = 13.
2. **Mutually exclusive?** No — a male nurse is both. So use Rule #2.
3. $P(N \cup M) = \frac{8}{13} + \frac{3}{13} - \frac{1}{13} = \frac{10}{13} \approx$ **0.769**

Sanity check by direct count: nurses-or-males = 7 female nurses + 1 male nurse + 2 male physicians = 10 people out of 13. ✔ Identical. Run this check whenever the numbers are small.

EXAMPLE 4-22 · DRIVING WHILE DEPRESSED

On New Year's night, the probability of a person driving while depressed is 0.32, the probability of a person having a driving accident is 0.09, and the probability of a person having a driving accident while depressed is 0.06. What is the probability of a person driving while depressed **or** having a driving accident?

1. **Extract:** P(D) = 0.32, P(A) = 0.09, P(D ∩ A) = 0.06.
2. **Mutually exclusive?** No — P(D ∩ A) = 0.06 ≠ 0, so they overlap. Rule #2.
3. $P(D \cup A) = 0.32 + 0.09 - 0.06 =$ **0.35**

TRAP — "ACCIDENT *WHILE* DEPRESSED" IS AN INTERSECTION, NOT A CONDITIONAL

The phrase "having a driving accident **while** depressed" means both things happen to the same person: $P(D \cap A) = 0.06$. It does *not* mean $P(A \mid D)$. If it had been conditional, the problem would say "*given that* the person is depressed" or "*of those who are depressed*". Misreading this turns 0.35 into a wrong answer, so slow down on the word that joins the two events.

3 · Extended addition rules (three or more events)

THREE MUTUALLY EXCLUSIVE EVENTS A, B, C

$$P(A \cup B \cup C) = P(A) + P(B) + P(C)$$

THREE EVENTS THAT ARE NOT MUTUALLY EXCLUSIVE

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P$$

Pattern: add the singles, subtract the pairs, add back the triple. The lecture notes point you to Exercises 19–21 for the mutually exclusive version and Exercise 24 for the general version.

متى أستخدم أي قاعدة؟

اسأل نفسك سؤالًا واحدًا: **هل يمكن للحدثين أن يقعا معًا؟**

- **لا** ← متنافيان (mutually exclusive) ← **P(A) + P(B)**
- **نعم** ← غير متنافيين ← **P(A) + P(B) − P(A∩B)**

وإذا ترددت، استخدم القاعدة الثانية دائمًا — فهي تعطي النتيجة الصحيحة في الحالتين لأن التقاطع يساوي صفرًا عند التنافي.

Chapter 4 study series — 11 files. Read the lessons in order; the revision files at the end draw on all of them.

00 · Exam brief — what the outline tests

01 · Lesson 4-1 — Sample Spaces and Probability

02 · Lesson 4-2 — The Addition Rules ← this file

03 · Lesson 4-3 — Multiplication Rules and Conditional Probability

04 · Lesson 4-4 — Counting Rules

05 · Lesson 4-5 — Probability and Counting Rules

06 · Master formula sheet

07 · Which rule do I use?

08 · The 14 traps that cost marks

09 · Review exercises, solved

10 · The A+ study protocol

Rebuilt from Dr. Ahmed Al-Hadhrami's lecture slides (Spring 24) and aligned to the SQU College of Science STAT 2101 course outline, Fall 2026. Textbook: Allan G. Bluman, *Elementary Statistics: A Step by Step Approach*, brief version, 10th edition, McGraw Hill (2017). A companion 100-question bank with full solutions accompanies this series.